

AJAY VENKATESH +1 516 585 9013 Email LinkedIn GitHub Portfolio		
Education		
New York University <i>Master of Science in Computer Engineering (GPA – 3.9/4)</i>		Sep 2024 – May 2026
• <i>Coursework:</i> System Design, Machine Learning, Deep Learning, Big Data, Advanced Java, Data Structures		
Skills		
Programming Languages: Python, Java, C++, TypeScript, JavaScript (ES6), SQL Web Technologies: REST API, GraphQL API, Microservices, React, Node.js, Express.js, Sequelize Database: RDBMS (MySQL, PostgreSQL), NoSQL (Dynamo DB, FireBase, MongoDB), Dataverse Cloud: AWS (AWS CDK, EC2, S3, Dynamo DB, API Gateway, Lambda, Cognito, IAM, Q, SageMaker), Azure (OpenAI services, Function Apps, Security Data lakes, Storage blobs) Tools: Git, Visual Studio, Cursor, Jupyter, Postman, Docker, Power Automate, Power BI, QuickSight AI/ML: PyTorch, Hugging Face, LLM Fine-tuning, RAG (LangChain, FAISS), NLP, SVM, Random Forest, MLflow, PySpark		
Experience		
CampusMesh		Jan 2026 – Present
Software Engineer <i>Real-time microservices, recommendation engine, and LLM-based AI retrieval pipelines</i> – Developed scalable Node.js microservices powering real-time academic workflows across study groups, messaging, calls, notifications, and profile management, backed by MySQL (Sequelize) and DynamoDB . – Deployed serverless architecture on AWS using Serverless Framework with Lambda, API Gateway, S3, and CloudFront across 4 environments (local, dev, staging, production); enforced security with KMS-managed keys, SSM Parameter Store , and least-privilege IAM . – Worked on LLM-based retrieval pipelines (RAG) for CampusIQ (AI Assistant), leveraging embedding-based semantic search and structured data indexing to improve contextual accuracy by 30% .		
Amazon		May 2025 – Aug 2025
Software Development Engineer Intern <i>Serverless AWS infrastructure, React 18 frontend, and Java backend systems</i> – Architected a secure UI console for Amazon fulfillment center teams to view image/video evidence across FCs; presented the system design for approval and established Role Based Access (4+ roles) using OAuth, Midway SSO, and HELIS . – Provisioned serverless Infrastructure-as-Code using AWS CDK with API Gateway, Lambda, DynamoDB, S3, and IAM, reducing provisioning time by 70% and enabling multi-stage, multi-region CI/CD deployments . – Built a React 18 frontend with optimized API integration, advanced filtering, client-side caching, reducing page load time by 40% . – Engineered backend services in Java using Coral, Smithy, Dagger , and CBOR serialization ; applied low-level design and design patterns reduced backend execution time by 35% and ensure reliability by unit and integration testing using Mockito .		
New York University (Novel AI Technologies, Inc.)		Aug 2024 – Present
Software Developer (On-Campus) <i>NSF-funded AI health platform, ETL pipelines, and full-stack insurance analytics</i> – Architected an NSF-funded (\$100K) AI-powered health app for iOS and Android , processing real-time wearable data; deployed cloud-hosted AI models for health matrix analysis and process food images. – Engineered ETL pipelines transforming NoSQL into PostgreSQL using concurrency control and locking mechanisms ; processed 250K+ records and powered AWS QuickSight dashboards for real-time health analytics. – Developed an insurance analytics platform (React, Node.js, REST APIs); deployed (Docker containers on EC2 , implemented Stripe billing, RBAC, and row-level security, generating \$50K+ revenue. – Streamlined underwriting via broker portal integrated with insurer platform; implemented document and media processing and WebSocket-based real-time communication ; secured 6 insurance clients .		
PricewaterhouseCoopers (PwC)		Aug 2021 – Aug 2024
Senior Software Engineer <i>Cloud-native backend, NLP/OCR automation, and data engineering on Azure</i> – Developed a distributed backend on Microsoft Azure for a health platform, integrating with Microsoft CRM and implementing event-driven processing and fault tolerance. – Led code reviews and mentored engineers on software design principles , reducing production bugs by 25% . – Deployed an NLP based OCR neural model on Azure to automate structured data extraction, reducing data processing time by 75% . – Optimized data ingestion with parallel processing for 100K-row from Excel; reduced execution time from 5 hours to 1 hour . – Built Power BI dashboards processing large datasets from multiple databases, improving reporting efficiency by 40% .		
Projects		
Personal Finance Analyzer (Python Data Analysis Project) <i>Python, Pandas, NumPy, Matplotlib, Seaborn</i> – Built a data-driven application to analyze personal financial transactions and categorize expenses using Pandas for efficient data processing. – Generated interactive visualizations and spending insights using Matplotlib and Seaborn to identify trends and optimize budgeting. – Designed reusable and modular code structure following OOP principles to improve maintainability and scalability.		
Certifications		
AWS Cloud Practitioner	Azure Data Fundamentals	Microsoft Solution Architect Expert
Power BI Data Analyst Associate	Microsoft Developer Associate	Stanford Machine Learning (Coursera)